EPSR-test-ex7 Empirical-potential reset (ireset 2)

Program(s): epsr | chained passes: 3 | verification: ALL IDENTICAL (all 3 passes)

What this case tests

Exercises `ireset 2` of the `epsr` engine [1,2]: the empirical potential is set to zero and any repulsive-core amplitudes `pcoefex` are cleared, while the reference potential, the EP amplitude `ereq` and the configuration are kept. Users take this route to restart a refinement without losing the equilibrated box [3].

Code exercised

Fortran (original, fortran_src/):

- `read_and_save_files.f`: `read_pcof` (EP coefficients 2..n set to 0, `pcoefex` := 0 for `ireset` = 2)
- `epsr_standalone_rev3.f`: `runepsr17` (`ireset` = 2 branch, reset after the run), `inita` (accumulators restarted), `generate_ep`, `ptgnsq`
- `read_and_save_files.f`: `save_files_others`

Rust (port, rust/epsr-core/src/):

- `pcof.rs`: `read_pcof`
- `epsr.rs`: `Epsr::setup`, `Epsr::iterate`; `structure.rs`: `State::inita`
- `potential.rs`: `generate_ep`, `ptgnsq`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. The EP is rebuilt from zero during the three runs at `ereq` = 10 kJ/mol. Unlike `ireset 1`, the `.pcof` pair table, `ereq` and the automatic-control settings are kept, and the leading coefficient of each pair is not reset.

Inputs

Copy of EPSR-test with only `ireset 2` (instead of 0) in `200DMSO400Water.EPSR.inp`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 3 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 3 passes.
Three chained passes; the Fortran writes `ireset 0` after pass 1, so passes 2-3 continue normally from the reset potential. In the Fortran reference `ereq` stays 10, the `.erg` history restarts (3 lines after pass 3) and new EP coefficients are refined from zero. All 27 output files are identical after every pass.

Notes

All `pcoefex` are already 0 in EPSR-test, so the clearing of the repulsive cores is executed but has no visible effect in this case.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.